

GUDIMELLA KRISHI KUMAR

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NIT Warangal, Telangana - 506004

 LinkedIn Profile |  GitHub Profile

EDUCATION

National Institute of Technology, Warangal
B.Tech. in Metallurgical and Materials Engineering
Minor in Computer Science and Engineering (CSE)

2023 - Present
CGPA: 9.16/10.0

Vision Junior College, Warangal (Intermediate)
Board of Intermediate Education, State

2021 - 2023
Percentage: 97.5%

TECHNICAL STRENGTHS

Modeling & Simulation	Phase Field Modeling (PFM), MicroSim
Programming & Tools	C++, Python (Scripting), Octave, Bash, LaTeX, Jupyter lab
Materials & Areas	Li-Polymer Batteries, Alloy Solidification, High temperature materials

PROJECTS & RESEARCH

Li-Polymer Battery Prototype (Current) Nov 2025 - Present
Personal Project and also as a part of Prototype club

- Developing a **prototype Li-polymer battery** focused on increasing the **safe operating temperature range**, critical for aerospace applications (drones, satellites) where thermal management is challenging.
- Research involves optimizing electrolyte composition and electrode material selection to enhance thermal stability and power density.

Phase Field Modeling of Rapid Alloy Solidification May 2025 - July 2025
Project Intern, IISc Bengaluru

- Developed and implemented a custom C++ code for the Phase Field Model (based on Karma's Mathematical Model) to simulate micro structural evolution during rapid solidification.
- Utilized **Finite Difference Methods (FDM)** for numerical discretizations, demonstrating competence in computational materials engineering.

Computational Tools for Experimental Works March 2025
Personal Project NIT Warangal

- Developed Python/C++ scripts to automate and replace time-consuming manual data reduction, significantly improving the efficiency and accuracy of experimental analysis.
- Tool 1 (Statistical Analysis): Implemented a statistical analysis code for Sieve Analysis data, relevant for characterizing .
- Tool 2 (ODE Solver): Created a numerical solver for **Ordinary Differential Equations (ODEs)** , .

Reclamation of Valuable Materials from War Waste October 2024
B.Tech minor case study Project for course

- Studied the extraction of Neodymium (Nd) from NdFeB magnets from scrap using (GMETs) and performed the cost analysis

WORKSHOPS AND TECHNICAL LEADERSHIP

Material Informatics Workshop - Technozion '25

Oct 2025

Main Coordinator

NIT Warangal

- Served as the ****Main Coordinator**** for a major Material Informatics workshop, managing logistics, curriculum, and participant engagement.
- Demonstrated organizational and technical leadership in promoting the use of Data Science (Informatics) for materials discovery.

ASM Workshop

June - July 2025

Participant

IISc Bengaluru

- Attended the ****ASM Workshop: Metallurgy for Practicing Engineers**** at IISc, focusing on industrial applications.
- had the opportunity to know about Coating technologies, Additive Manufacturing, Aerospace heat treatment and its norms

MicroSim Workshop

June - July 2025

Participant

IISc Bengaluru

- Completed the ****5-day MicroSim Workshop**** (Microstructure Simulation) at IISc Challakere, enhancing skills in advanced computational modeling techniques.
- Had the opportunity to know and interact with other people involved in phase field Modeling in India.

RELEVANT COURSEWORK

Computational & Advanced Materials

Data Structures and Algorithms

AI and ML in Material Science

Material Design for Versatile Applications

Electronic and Magnetic Materials

Core Engineering

Transport Phenomena

Mechanical Behavior of Materials

Engineering Mechanics

Physical Metallurgy & Phase Transformation

LEADERSHIP AND ACHIEVEMENTS

Leadership Positions

2024 - Present

- Additional Secretary, Metallurgical and Materials Engineering Student Society (*3 months and continuing*).
- Executive Member, Prototype Club, involved in hands-on technical project development.

Academic Achievements

2023 - 2025

****Merit Student:**** Consistently among the ****top 3 students**** in the class for two academic years (2023-24, 2024-25). ****Academic Excellence:** Secured admission to NIT Warangal, ranking among the ****Top 1%**** of candidates in JEE Mains 2023.

EXTRA ACADEMIC ACTIVITIES

Assisted my friend Shri Dumpala Suchethan Reddy for his First Video on our College;

Acted as Rajguru as a part of Bhagath Singh Drama in an college level event